

# UTSAV CHAUDHARY

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Senior **DevOps Engineer** with **5+ years** supporting and delivering **production-grade AWS and Azure infrastructure** for high availability systems in regulated and enterprise environments. Proficient in **Kubernetes and Helm orchestration**, **Infrastructure as Code** with **Terraform** and **Ansible**, and **multicloud deployments** across Dev, Test, UAT, QA, and Production. Experienced contributing to **disaster recovery**, **observability**, and compliance-ready platform operations that meet **RTO** and **RPO** targets for business-critical systems.

## Technologies

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**Cloud & Infrastructure:** AWS (EC2, EKS, ECS, Fargate, Lambda, RDS, S3, EBS, VPC, IAM, CloudWatch, API Gateway, Secrets Manager, Route 53, ALB/ELB, CodeDeploy, KMS, CloudHSM), Azure (App Services, AKS, Azure DevOps, Azure SQL, Application Gateway, Front Door, Key Vault, Entra ID), Docker, Kubernetes, Helm, Rancher, Argo CD

**IaC & Automation:** Terraform, Ansible, Puppet, Chef, Bash, Python, PowerShell, YAML, CloudFormation, Nginx, Cron

**CI/CD & Version Control:** Jenkins, GitHub Actions, Bitbucket Pipelines, AWS CodePipeline, Azure DevOps, Git, GitHub, GitLab, Bitbucket, AWS CodeArtifact

**Monitoring & Logging:** Prometheus, Grafana, Elasticsearch, Logstash, Kibana, CloudWatch, Application Insights, Datadog, Elastic Stack

**Networking & Security:** VPCs, VNets, Security Groups, NSGs, IAM Policies, DNS, VPN, LDAP, RBAC, SSL/TLS, OAuth 2.0, HIPAA, GxP

**Development & Tools:** Python, TypeScript, JavaScript, Groovy, FastAPI, Node.js, React.js, Next.js, REST APIs, gRPC, WebSockets, PostgreSQL, MongoDB, Redis, Firebase, Kafka, Jira, Confluence

**Operating Systems:** Linux (Ubuntu, CentOS, RHEL), Windows Server, macOS, Raspberry Pi

## Experience

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**Sr. DevOps Engineer**, Hybrid, **Labs196**, TX

Feb 2025 - Present

- Collaborated with platform and compliance teams to support a **multicloud strategy** across **AWS** and **Azure**, helping deploy and maintain **Kubernetes** workloads on **EKS** and **AKS** with **Helm** across **Dev**, **Test**, **UAT**, **QA**, and **Production** for regulated lab and research platform services.
- Contributed to **GitOps** release workflows using **Argo CD**, collaborating with platform and staff engineers to standardize application deployments and reduce manual release steps across **GxP-validated** environments.
- Supported centralized cluster operations with **Rancher** managing **EKS** and **AKS** clusters, integrating **Azure Entra ID** for **RBAC**, least-privilege access, and audit-ready logging aligned with **FDA** and internal compliance reviews.
- Maintained multicloud infrastructure through **Terraform** and **Ansible**, managing **AWS** resources including **VPCs**, **security groups**, **IAM policies**, and **RDS**, and **Azure** resources including **VNets**, **NSGs**, **Key Vault**, and **Azure SQL** with version-controlled **IaC**.
- Contributed to cross-cloud **observability** using **Prometheus**, **CloudWatch Logs**, and the **ELK Stack**, supporting on-call incident response and production **SLA** targets for platform and data pipeline services.
- Helped automate backup and retention workflows for **S3**, **Azure Blob Storage**, and cloud-hosted databases, following **GxP** retention requirements established by compliance and platform leadership.
- Participated in **on-call** rotations and **incident response**, creating **runbooks** and coordinating with engineering teams to troubleshoot and restore production services across both cloud environments.

**DevOps Engineer**, On-Site, **Delta Dental**, GA

Jan 2024 - Feb 2025

- Collaborated with a distributed platform team, including **California-based engineers**, to support production **Azure Kubernetes Service (AKS)** clusters running **containerized claims and payor workloads** behind **Azure Application Gateway**, helping maintain **99.9% availability** for **member eligibility** and **claims processing** services.
- Contributed to centralized **monitoring and alerting** using **Application Insights**, **Azure Monitor**, and **Grafana dashboards**, integrated with **Azure Front Door** health probes and gateway metrics, supporting a team reduction in **mean time to detect (MTTD)** from **30 minutes** to **5 minutes** across **claims adjudication** and **enrollment** workflows.
- Assisted in migrating legacy applications to **cloud-native AKS** on Azure, working with senior DevOps engineers to improve **resource utilization** by **40%** for the **member enrollment** and **provider portal** platforms.
- Supported **canary release** rollouts through **Azure DevOps** pipelines with **Application Gateway** traffic splitting under senior guidance, contributing to a **75%** reduction in deployment-related incidents for the **member self-service portal**.
- Helped automate **backup** and **disaster recovery (DR)** for **Azure SQL** and cloud-hosted data services, supporting team targets of **RPO 10 minutes** and **RTO 30 minutes** for critical **claims transaction processing** systems.

- Supported **HIPAA-aligned** security practices using **Azure Key Vault** for encryption and key rotation of **member PHI and claims data** at rest and in transit, following standards set by senior engineers and the compliance team.

**DevOps Engineer CO-OP**, On-Site, **Midh Technologies**, TX

June 2023 - Dec 2023

- Managed production infrastructure for ML workloads, supporting **GPU backed Kubernetes clusters** on **AWS EKS** with cost optimization strategies including node pool configuration, resource quotas, and auto scaling policies.
- Designed high-performance inference infrastructure deploying **FastAPI** endpoints behind **AWS API Gateway** with **IAM authentication**, request throttling, and **Lambda** integrations for scalable ML inference.
- Developed infrastructure automation through **Ansible playbook** development, reducing manual server configuration and deployment time by **50%** while ensuring configuration consistency.
- Built production **CI/CD pipelines** with **GitHub Actions**, automating testing, Docker image builds, security scanning, and deployments to **AWS Amplify** and **EKS** environments.
- Implemented infrastructure security practices including **RBAC** policies, **IAM role** management, and **VPC** security configurations across AWS services and application layers.

**DevOps/Platform Engineer**, On-Site, **Compendious Medialabs Pvt. Ltd.**, IN

June 2021 - July 2022

- Managed hybrid infrastructure for **on premise Linux servers** (Ubuntu) with containerized deployments using **Docker**, establishing infrastructure reliability and automation through **Bash** scripting and configuration management.
- Designed and implemented self hosted CI/CD infrastructure deploying **GitHub Actions runners** on local infrastructure, building production grade pipelines for automated testing, Docker builds, and multiple environment deployments.
- Established production grade reverse proxy infrastructure with **Nginx** and **SSL/TLS** termination, optimizing external service exposure with **250ms** average response time and high availability.
- Implemented observability strategy deploying **Prometheus** and **Grafana** monitoring on local infrastructure; automated infrastructure tasks with **Python/Bash** scripting, reducing manual operational overhead by **60%**.

**Software Engineer / DevOps**, Remote, **VRoKCs**, MO

March 2020 - Sept 2020

- Managed production infrastructure on **Azure**, deploying **Flask web applications** on **Azure App Services** with high availability configurations, auto scaling, and integrated monitoring.
- Built infrastructure automation through **Azure DevOps** pipelines for ML model deployment, implementing artifact storage with **Azure Blob Storage** and version control workflows.
- Designed containerized infrastructure using **Azure Container Instances** and **Docker** for ML inference services, ensuring environment parity between development and production with consistent deployment practices.
- Developed CI/CD infrastructure using **Azure DevOps** for ML lifecycle management, automating training and deployment workflows and reducing deployment cycles by **40%** through pipeline optimization.

**Web/Game Developer**, On-Site, **underDogs Gaming Studios**, IN

March 2019 - Aug 2019

- Contributed to backend API infrastructure for gaming services, integrating and optimizing **Quarters API** across multiple games with secure transaction, supporting **load balancing** and **API rate limiting** for high availability systems.
- Supported infrastructure reliability initiatives by optimizing real time client server communication, validating and refining backend interaction flows using **Python** and **C#**, implementing error handling and health monitoring.
- Developed unified transaction infrastructure connecting external services (**payments, inventory, shipping**) to internal game systems, following consistent API patterns and infrastructure best practices.
- Participated in infrastructure security and testing practices, implementing **API security best practices**, comprehensive logging, and integration testing frameworks for edge cases including payment failures and network resilience.

## Education

**State University of New York at Binghamton**

Master of Science in Computer Science

Aug 2022 - Dec 2023

3.44/4.00 GPA

## Certifications



**Microsoft Certified: DevOps Engineer Expert**

AZ-400: Designing and Implementing Microsoft DevOps Solutions

AZ-900: Microsoft Certified: Azure Fundamentals

**Google IT Automation with Python** (Specialization)

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